

KINEXUS'25

INTER IIT NIT HACKATHON

Group Name: Smooth Operators

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Problem Statement 2: Cost-Effective Alternative Catalysts for Industrial Hydrogenation Process

Team Members:

Boggarapu Harshith

Arkadip Ghara

Om prakash Meghwal

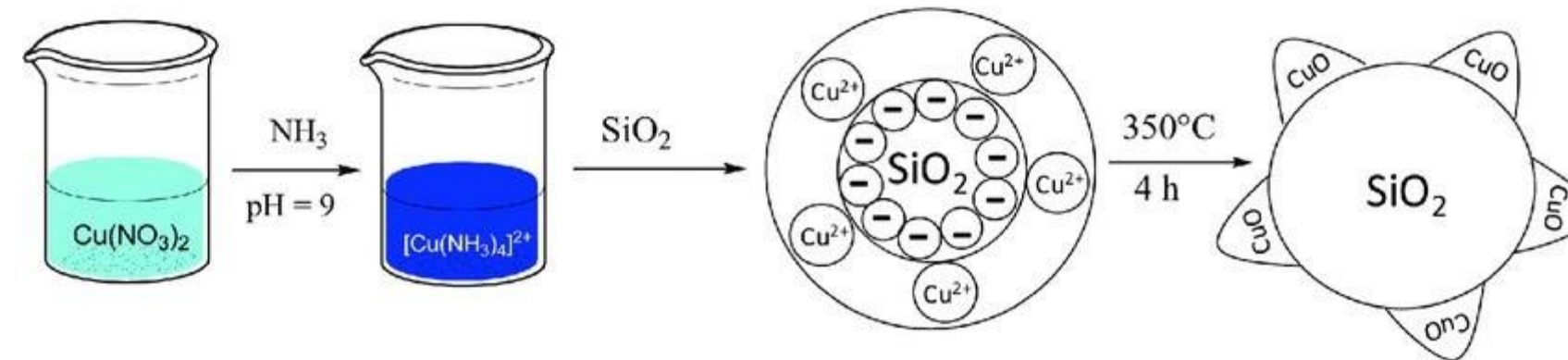
Tanmay Kumar

Ayush Raj



Catalyst Overview

- **Catalyst Selection:** To address the environmental and economic limitations of traditional Nickel systems, the selection of **8% Cu/SiO₂** prepared via the **Chemisorption - Hydrolysis (CH)** method provides a high-activity, chromium-free alternative for the partial hydrogenation of fatty acids **linolenic(C 18:3)** and **linoleic(C 18:2)** acids.
- **Catalyst Preparation:** To prepare the Cu/SiO₂ catalyst using the Chemisorption hydrolysis (CH) method, the silica support is added to a **copper-ammonia complex** solution followed by slow dilution with water to trigger precise hydrolysis and deposition. For the **ex situ activation**, the catalyst is pre-reduced under hydrogen at **270°C** in the absence of oil, ensuring that any water formed is removed to maintain maximum activity.

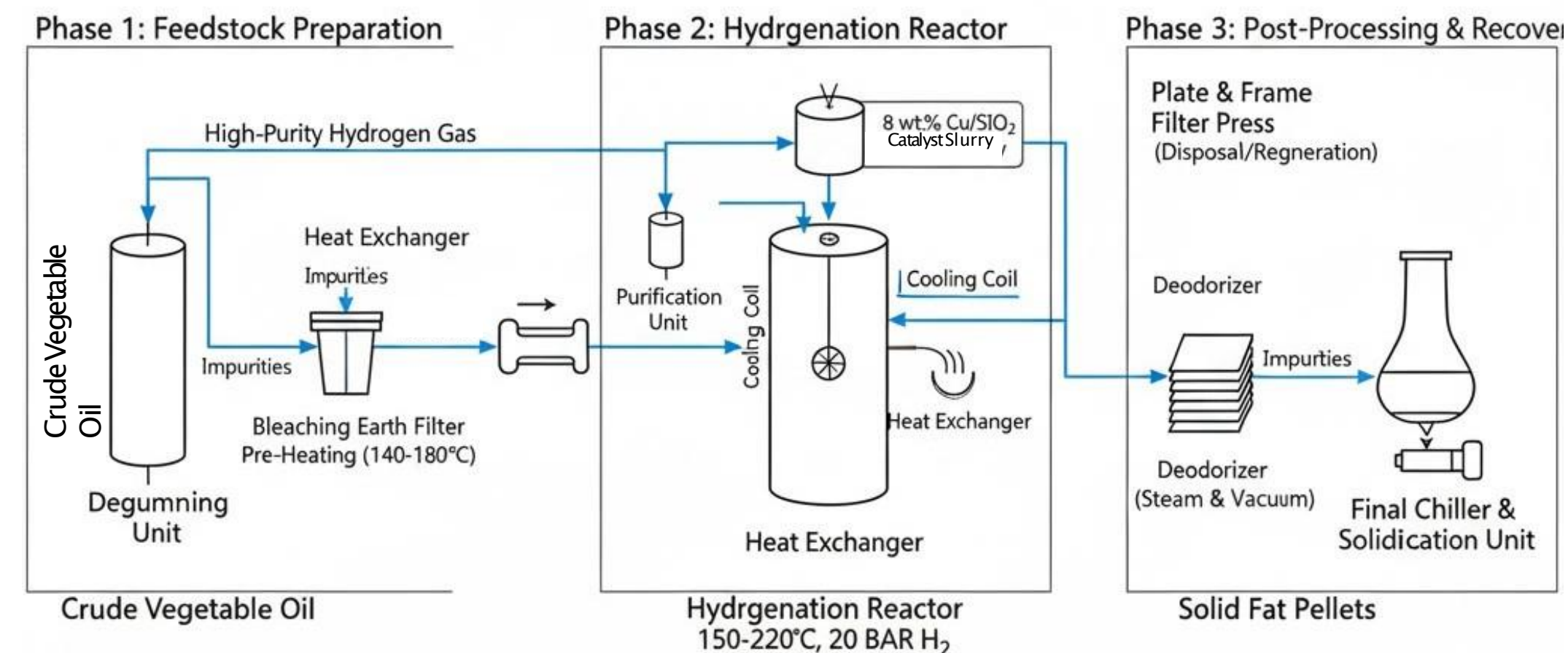


- **Catalyst Deactivation and Stability:** Industrial catalysts typically deactivate through **poisoning** by impurities, **fouling (coking)**, or **sintering** of metal particles. While standard Nickel catalysts are highly sensitive to these contaminants, Cu/SiO₂ prepared by the CH method exhibits unusual stability. This robustness is achieved through **ex situ pre-reduction at 270°C**, which **removes all water** formed during synthesis. By eliminating this water which would otherwise trap in the pores and limit activity the catalyst maintains a stable, high-density surface of active sites for long-term industrial use.
- **Activity and Selectivity Trade-Offs:** In the industrial hydrogenation of fatty acids, activity and selectivity trade-offs involve balancing the high reaction speed enabled by Cu/SiO₂'s unique structure against the thermodynamic challenge of reaching full saturation. While the chemisorption-hydrolysis (CH) method creates **3.5 nm copper** nanoparticles with sites that are **100 times more active** for hydrogen dissociation than traditional catalysts.
- **Scalability:** The process is highly adaptable across various abundant, commercial silica supports and maintains turnover frequencies up to **100 times higher** than traditional penetration methods, ensuring the high throughput required for large-scale production. Furthermore, the robust nature of the CH-prepared copper phase allows it to resist **deactivation from water** and remain active over multiple reuse cycles.
- **Safety and Handling:** Supported copper catalysts are generally considered safer for industrial environments than traditional nickel alternatives because they are **non-pyrophoric**, meaning they do not spontaneously **ignite when exposed to air**, and they provide an environmentally friendly, **chromium-free** system that simplifies both personnel handling and waste compliance.
- **Less waste generation per MT:** By utilizing a significantly lower metal loading of **8 wt.% Cu compared to the 22 wt.%** typically found in commercial **Nickel catalysts**, the **Cu/SiO₂** system generates substantially less spent catalyst **waste per metric ton (MT)** of processed oil. This efficiency is further enhanced by the catalyst's high stability and reusability, which reduces the frequency of catalyst disposal and the associated environmental compliance costs.

Process Design

- Feedstock:** The feedstock primarily consists of vegetable oils (such as rapeseed, soyabean, and sunflower) and their methylesters, which are composed of unsaturated fatty acids like **linolenic (C18:3)**, **linoleic (C18:2)**, and **oleic (C18:1)** acids that require selective hydrogenation to improve **oxidation stability** and **viscosity** for use as environmentally friendly lubricants or biodiesel.
- Reactor Type:** The process utilizes a **slurry-phase batch reactor**, where the supported copper catalyst is added directly to the oil to form a slurry that is kept in suspension through **constant agitation** using a magnetic stirring head and stirring rod during the hydrogenation process.
- Hydrogenation Type:** The process prefers **partial (selective) hydrogenation** because it allows for the complete elimination of the unstable **trienic component (C18:3)** and a significant reduction of the **dienic component (C18:2)** to improve oxidative stability, while crucially avoiding full hydrogenation to stearic acid (C18:0) to maintain the oil's fluidity and low-temperature properties (pour point) required for **lubricants**.
- System Variables:** The system variables involve maintaining a reaction temperature of 180°C and a hydrogen pressure of 6 to 20 bar, while the catalyst is prepared through drying at 120°C, **calcination at 350°C**, and an ex situ **pre-reduction** treatment at 270°C.
- Parameters for complete saturation :** Complete saturation to stearic acid requires **high-temperature (200–250 °C)** and **high-pressure hydrogenation (20–50 bar H₂)** to enable hydrogenation of mono-unsaturated fatty acids using **Cu/SiO₂ catalyst**. The reaction is typically maintained for **3–8 hours** to ensure sequential conversion of C18:3 and C18:2 through C18:1 to C18:0.

Process Flow Diagram



- Feedstock Preparation** – Crude vegetable oil undergoes degumming and is passed through a bleaching earth filter at temperatures between **140°C and 180°C** to remove **phospholipids and impurities** before entering the reactor.
- Hydrogenation Reactor** – The pre-treated oil is reacted with high-purity hydrogen gas at **20 bar and 150-220°C**. An **8 wt.% Cu/SiO₂** catalyst slurry is introduced to facilitate the chemical saturation of the fatty acids, with cooling coils managing the heat of the reaction.
- Post-Processing & Recovery** – The hydrogenated product is filtered via a **plate and frame press** to remove the **catalyst**, processed through a steam and vacuum deodorizer to eliminate odors, and finally cooled in a chiller to produce solidified fat pellets.

Performance and Economic Analysis

Category	Nickel catalyst	Supported copper Catalyst
Cost Estimation	Nickel is more expensive and often requires modification to control excessive activity.	Uses copper, which is cheaper and more abundant; no need for costly additives or alloying.
Catalyst Efficiency	High activity but poor selectivity; hydrogenates C18:3, C18:2, and C18:1, leading to over-hydrogenation.	Highly efficient and selective; hydrogenates C18:3 and C18:2 while preserving C18:1; produces high-oleic oils.
Selectivity Control	Excellent control over hydrogenation depth; minimal Stearic acid formation.	Excellent control over hydrogenation depth; minimal Stearic acid(C18:0) formation.
Deactivation	Faster deactivation due to carbon deposition, strong adsorption, and surface poisoning.	Lower deactivation rate due to ex-situ pre-reduction and removal of water; stable Cu ⁰ active sites.
Cold flow properties	Poor cold properties ; high pour point(+7°C)	Excellent cold properties; low pour point(-12 to -15°C)
Sustainability	Generate high saturated and trans-fat content; less suitable for green applications	Chromium-free, environmentally safer; enables biodegradable and ecofriendly lubricants.
Scalability	Requires strict control to prevent over hydrogenation; complicates large scale operation	Operates efficiently at moderate temperature(~180°C) and pressure(6–20 bar);easier industrial scale up.
Product Quality	Produces oils with high saturation, poor fluidity, and inferior lubricant quality.	Produces oils with high oleic content, good viscosity, oxidation stability, fluidity.

Research and references

- **ScienceDirect (Research Article):** Environmentally friendly lubricants through selective hydrogenation of rapeseed oil over supported copper catalyst
- **Catalyst Synthesis Study:** High-resolution imagery and data on Supported Copper Catalysts via Chemisorption-Hydrolysis.
- **Industrial Benchmarking:** Comparison data for Finely Divided Nickel (Ni) catalysts including cost, activity, and poisoning metrics.
- **Process Flow Diagram (PFD):** AI-generated architectural map using Gemini (Nano Banana) based on specific hydrogenation parameters.

<https://www.sciencedirect.com/science/article/pii/S0926860X0200128X>

https://www.researchgate.net/figure/Formation-of-copper-oxide-nanoparticles-over-silica-by-means-of-the_fig1_233838610

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